Cloud Computing – I UNIT

2 marks:

1. Define cloud computing?

Cloud computing is a technology that delivers computing services including servers, storage, databases, networking, and software over the internet. Users can access and use these resources on-demand, paying only for what they use, without the need to own or maintain physical hardware. This approach offers flexibility, scalability, and cost-efficiency for individuals and organizations.

2. List the different computing paradigms?

- Parallel Computing
- Distributed computing
- Grid Computing
- Cloud Computing

3. What do you mean by parallel computing?

Parallel computing refers to the process of executing several processors an application or computation simultaneously. Here, the large problems break into independent, smaller, usually similar parts that can be processed in one go. It is done by multiple CPUs communicating via shared memory. It helps in performing large computations as it divides the large problem between more than one processor.

4. What do you mean by distributed computing?

Distributed computing refers to a system where processing and data storage is distributed across multiple devices or systems, rather than being handled by a single central device. Each device or system has its own processing capabilities and may also store and manage its own data. These devices or systems work together to perform tasks and share resources, with no single device serving as the central hub.

5. Give any two advantages of distributed computing?

***Performance**:* It helps to improve performance by having each computer in a cluster handle different parts of a task simultaneously.
***Cost-effectiveness**:* Distributed computing can use low-cost, off-the-shelf hardware.

6. What do you mean by cluster computing?

A cluster computing system consists of a set of the same or similar type of processor machines connected using a dedicated network infrastructure. These interconnected processor machines, referred to as nodes or cluster nodes, work together as a single system to perform computation tasks. All processor machines share the resources. The nodes need to communicate with one another to work cooperatively and meaningfully together to solve the problem at hand.

7. What is the use of grid computing?

Grid computing is used to perform large tasks or solve complex problems that would be difficult to do on a single computer. For example, meteorologists use grid computing for weather modelling, analyzing data from telescopes and satellites etc.

8. What are the different characteristics of cloud computing?

• No up-front commitments • On-demand access • Nice pricing • Scalability • Elasticity

9. Expand CSCs.

Cloud Service Commerce

10. Expand CSPs.

Cloud Service Provider

11. Compare cloud computing and parallel computing.

Cloud computing allows us to access and use different services over the Internet, like databases, storage, servers, and software. In parallel computing, a big task is split into smaller parts and each part is processed simultaneously by multiple processors in a single computer or a group of computers.

12. Compare parallel computing and distributed computing.

Parallel computing typically requires one computer with multiple processors. Distributed computing, on the other hand, involves several autonomous (and often geographically separate and/or distant) computer systems working on divided tasks.

13. Differentiate between cluster computing and grid computing.

Cluster computing involves a tightly knit group of computers, typically located close together and connected by a high-speed network. These computers share a common storage system and operate under a single centralized management system.

Grid computing involves geographically dispersed computers that may belong to different organizations. These computers are typically underutilized and connected through a standard network like the internet. Resources are managed in a decentralized manner.

14. What is the advantage and disadvantage of cloud computing?

Advantages:

Back-up and restore data: Once the data is stored in the cloud, it is easier to get back-up and restore that data using the cloud.
Improved collaboration: It allows groups of people to quickly and easily share information in the cloud via shared storage.

Disadvantages:

Internet Connectivity: If you do not have good internet connectivity, you cannot access the data. However, we have no any other way to access data from the cloud.
Vendor lock-in: As different vendors provide different platforms, that can cause difficulty moving from one cloud to another.

15. What is the disadvantage of cloud computing?

Refer above answer

16. List the various applications of cloud computing.

Cloud Computing in Education: CC in the education sector brings an unbelievable change in learning by providing e-learning, online distance learning platforms and student information portals to the students. Eg: Google Meets.

Cloud Computing in Medical Fields: CC is used for storing and accessing the data through the internet without worrying about any physical setup. It facilitates easier access and distribution of information among the various medical professional and the individual patients.

17.Mention the different trends in cloud computing?

• AI and ML powered cloud • Edge Computing • Serverless Computing • Blockchain • IoT

18.List the leading platforms of cloud computing.

Amazon web services (AWS) Google Cloud Platform (GCP)

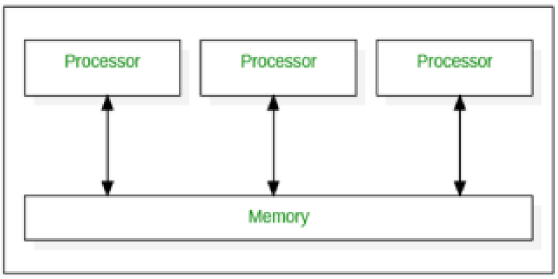
19.Expand AWS and GCP

AWS - Amazon web services GCP - Google Cloud Platform

Long answer questions:

1. Explain in detail about parallel computing with example.

Parallel computing refers to the process of executing several processors an application or computation simultaneously. Here, the large problems break into independent, smaller, usually similar parts that can be processed in one go. It is done by multiple CPUs communicating via shared memory. It helps in performing large computations as it divides the large problem between more than one processor.



Examples: The iPhone 5 has a 1.5 GHz dual-core processor. The iPhone 11 has 6 cores. The Samsung Galaxy Note 10 has 8 cores. These phones are all examples of parallel computing.

Advantages:

• Parallel computing saves time by allowing programs to be executed in less time.
• Larger problems must be solved in a shorter period.

Limitations:

• The multi-core architectures consume high power consumption.
• It addresses Parallel architecture that can be difficult to achieve.

2. Explain in detail distributed computing with examples.

Distributed computing refers to a system where processing and data storage is distributed across multiple devices or systems, rather than being handled by a single central device. Each device or system has its own processing capabilities and may also store and manage its own data. These devices or systems work together to perform tasks and share resources, with no single device serving as the central hub.

One example of a distributed computing system is a cloud computing system, where resources such as computing power, storage, and networking are delivered over the Internet and accessed on demand. In this type of system, users can access and use shared resources through a web browser or other client software.

Advantages:

***Performance**:* It helps to improve performance by having each computer in a cluster handle different parts of a task simultaneously.
***Cost-effectiveness**:* Distributed computing can use low-cost, off-the-shelf hardware.

Disadvantages:

Complexity: These systems can be more complex as they involve multiple devices or systems that need to be coordinated and managed.
Security: It can be more challenging to secure a distributed system, as security measures must be implemented on each device or system to ensure the security of the entire system.

3. Explain briefly about cluster computing and grid computing with examples.

A cluster computing system consists of a set of the same or similar type of processor machines connected using a dedicated network infrastructure. These interconnected processor machines, referred to as nodes or cluster nodes, work together as a single system to perform computation tasks. All processor machines share the resources. The nodes need to communicate with one another to work cooperatively and meaningfully together to solve the problem at hand.

Eg: Imagine a company that needs to process large volumes of data for various analytical tasks, such as data mining, data warehousing, or generating reports. Instead of relying on a single, powerful computer, they use a cluster of computers for data processing.

Grid computing is a distributed architecture of multiple computers connected by networks to accomplish a joint task. These tasks are compute-intensive and difficult for a single machine to handle. Several machines on a network collaborate under a common protocol and work as a single virtual supercomputer to get complex tasks done.

Eg: Meteorologists use grid computing for weather modelling, analyzing data from telescopes and satellites etc.

4. What are the different computing paradigms? Explain briefly.

Cloud Computing: Cloud computing is a technology that delivers computing services including servers, storage, databases, networking, and software over the internet. Users can access and use these resources on-demand, paying only for what they use, without the need to own or maintain physical hardware. This approach offers flexibility, scalability, and cost-efficiency for individuals and organizations.

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5. How does Grid computing works?

A typical grid computing network consists of three machine types:

Control node/server: A control node is a server or a group of servers that administers the entire network and maintains the record for resources in a network pool.

Provider/grid node: A provider or grid node is a computer that contributes its resources to the network resource pool.

User: A user refers to the computer that uses the resources on the network to complete the task.

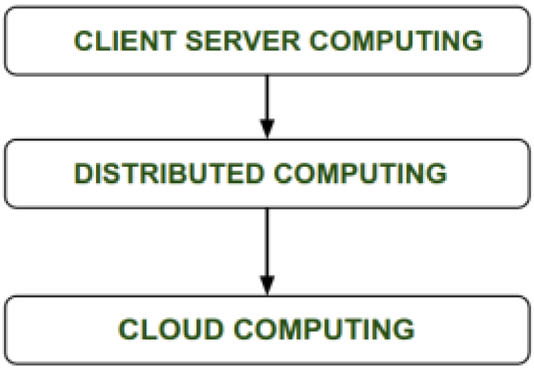
Grid computing operates by running specialized software on every computer involved in the grid network. The software coordinates and manages all the tasks of the grid. Fundamentally, the software segregates the main task into subtasks and assigns the subtasks to each computer. This allows all the computers to work simultaneously on their respective subtasks. Upon completion of the subtasks, the outputs of all computers are aggregated to complete the larger main task.

6. Compare various computing technologies.

Task	Parallel Computing	Distributed computing	Cluster computing	Grid Computing	Cloud Computing
Task Execution	Concurrent processing	Distributed processing	Concurrent processing	Distributed processing	On-demand resource allocation
Use case	Scientific simulations	Collaborations	HPC,Data analysis	Collaborative research	Versatile application
Architecture	Shared/ distributed memory	Networked computers	Cluster of computers	Geographically dispersed resources	Virtualized infrastructure
Scalability	Limited	Scalable	Highly scalable	Highly scalable	Highly scalable
Resource management	Manual	Varies by approach	Managed by middleware	Managed by middleware	Managed by cloud provider
Examples	Supercomputers, MPI	Hadoop, MPI, BOINC	Beowulf cluster	Open science grid, globus	AWS, AZUI, GCP

7. Write a note on history of cloud computing.

Before the advent of computing, client-server architecture was commonly used, where all data and control of clients resided on the server side. In this setup, when a single user needed to access data, they had to connect to the server first before gaining appropriate access. Subsequently, distributed computing came into existence. In distributed computing, all computers are interconnected, allowing users to share resources as needed. Nevertheless, distributed computing also had its limitations. To address the challenges faced in distributed systems, cloud computing emerged as a solution.



8. Explain the features of cloud computing.

• **No up-front commitments:** This means you don't have to contractually agree to a specific amount of usage up front. Many contracts have minimums and maximums built in, so even if you don't use a product much, you would still be obligated to a minimum payment

• **On-demand access:** The Cloud computing services does not require any human administrators, user themselves are able to provision, monitor and manage computing resources as needed.

• **Nice pricing :** Cloud providers offer a variety of pricing models, including pay-per-use, subscription-based, and spot pricing, allowing users to choose the option that best suits their needs.

• **Scalability:** Cloud scalability in cloud computing refers to the ability to increase or decrease IT resources as needed to meet changing demand.

• **Elasticity:** Cloud Elasticity is the property of a cloud to grow or shrink capacity for CPU, memory, and storage resources to adapt to the changing demands of an organization.

• **Efficient resource allocation:** The IT resource (e.g., networks, servers, storage, applications, and services) are shared across multiple applications efficiently when they require.

• ***Energy efficiency:*** Cloud computing is an internet based computing which provides metering based services to consumers. It means accessing data from a centralized pool of compute resources that can be ordered and consumed on demand. There are multiple techniques and algorithms used to minimize the energy consumption in cloud.

9. Explain advantages of cloud computing.

• **Back-up and restore data:** Once the data is stored in the cloud, it is easier to get back-up and restore that data using the cloud.

• **Improved collaboration:** It allows groups of people to quickly and easily share information in the cloud via shared storage.

• **Excellent accessibility:** Cloud allows us to quickly and easily access store information anywhere, anytime in the whole world, using an internet connection. An internet cloud infrastructure increases organization productivity and efficiency by ensuring that our data is always accessible.

• **Low maintenance cost:** Cloud computing reduces both hardware and software maintenance costs for organizations.

• **Mobility:** Cloud computing allows us to easily access all cloud data via mobile.

• **Services in the pay-per-use model:** Cloud computing offers Application Programming Interfaces (APIs) to the users for access services on the cloud and pays the charges as per the usage of service.

10.Explain disadvantages of cloud computing.

• **Internet Connectivity:** If you do not have good internet connectivity, you cannot access the data. However, we have no any other way to access data from the cloud.
• **Vendor lock-in:** As different vendors provide different platforms, that can cause difficulty moving from one cloud to another.

• **Limited Control:** As we know, cloud infrastructure is completely owned, managed, and monitored by the service provider, so the cloud users have less control over the function and execution of services within a cloud infrastructure.

• **Security:** Before adopting cloud technology, you should be aware that you will be sending all your organization's sensitive information to a third party, i.e., a cloud computing service provider. While sending the data on the cloud, there may be a chance that your organization's information is hacked by Hackers.

• **Lack of support staff:** Some cloud companies do not provide proper support to their clients; then, you have to only depend on FAQs or online help.

• **Technical issues:** Due to frequent version releases of some applications, you have to constantly upgrade your systems to meet a market need; in between these updates, there is a chance that you may be stuck on some technical problems

11.Explain various applications of cloud computing.

• **Cloud Computing in Education:** Cloud computing in the education sector brings an unbelievable change in learning by providing e-learning, online distance learning platforms and student information portals to the students. It is a new trend in education that provides an attractive environment for learning, teaching, experimenting, etc. to students, faculty members, and researchers. Everyone associated with the field can connect to the cloud of their organization and access data and information from there.

• **Cloud Computing in Medical Fields:** CC is used for storing and accessing the data through the internet without worrying about any physical setup. It facilitates easier access and distribution of information among the various medical professional and the individual patients.

• **Entertainment Applications:** Different types of entertainment industries reach near the target audience by adopting a multi-cloud strategy. Cloud-based entertainment provides various entertainment applications such as online music/video, online games and video conferencing, streaming services, etc. and it can reach any device be it TV, mobile, set-top box, or any other form.

• **E-commerce Application:** Users respond quickly to the market opportunities as well as the traditional e-commerce responds to the challenges quickly. Cloud-based e-commerce gives a new approach to doing business with the minimum amount as well as minimum time possible. Customer data, product data, and other operational systems are managed in cloud environments.

• **Anti-Virus Applications:** nowadays cloud computing provides cloud antivirus software which means the software is stored in the cloud and monitors your system/organization's system remotely. This antivirus software identifies the security risks and fixes them. Sometimes also they give a feature to download the software

• **Bigdata Analysis:** We know the volume of big data is so high where storing that in traditional data management system for an organization is impossible. But cloud computing has resolved that problem by allowing the organizations to store their large volume of data in cloud storage without worrying about physical storage.

12.Explain various trends in cloud computing.

• **AI and ML powered cloud:** Artificial Intelligence and Machine Learning are two technologies that are closely related to cloud computing. AI and ML services are more cost-effective since large amounts of computational power and storage space are needed for data collection and algorithm training. They are a solution for managing massive volumes of data to improve tech company productivity.

• **Edge Computing:** Edge Computing is one of the biggest trends in cloud computing. Here, data is stored, processed at the edge of the network, and analyzed geographically closer to its source. Faster processing and reduced latency can be achieved due to the increasing use of 5G. Edge computing has major benefits which include more privacy, faster data transmission, security, and increased efficiency.

• **Serverless Computing:** Here, compute resources are provided as a service rather than installed on physical servers. This means that the organization only pays for the resources they use rather than having to maintain its servers. In addition, serverless cloud solutions are becoming popular due to ease of use and ability to quickly build, deploy and scale cloud solutions.

• **Blockchain :** Blockchain is a linked list of blocks containing records and keeps growing as users add to it. Cryptography is used to store data in blocks. It offers excellent security, transparency, and decentralization. It can process vast amounts of data and exercise control over documents economically and securely.

• **IoT:** Internet of Things(IoT) is a well-known trend in cloud computing. It is a technology that maintains connections between computers, servers, and networks. It functions as a mediator and ensures successful communication and assists in data collection from remote devices.

• **Cloud Security and Resilience:** Cloud migration offers many benefits, efficiencies, and conveniences but presents several security risks. Because of it, investing in cyber security and building resilience against everything from data loss to the impact of a pandemic on global business will become a priority increasingly during the coming year.

13. Explain any most leading platforms of cloud computing.

• **Amazon web services (AWS) :** It is one of the most well-integrated and widely adopted cloud platforms in the world. AWS offers comprehensive cloud IaaS (Infrastructure as a Service) services ranging from virtual compute, storage, and networking to complete computing stacks. AWS is mostly known for its compute and storage-on-demand services.

• **Google Cloud Platform (GCP) :** GCP is a suite of cloud computing services that runs on the same infrastructure that Google uses internally for its end-user products, such as Google Search, Gmail, Google Drive, and YouTube. Google lists over 100 products under the Google Cloud brand.

• **Microsoft Azure :** Microsoft Azure is a cloud operating system and a platform for developing applications in the cloud. It provides a scalable runtime environment for Web applications and distributed applications. Azure facilitates the development, testing, deployment, and 26 management of services and applications through data centers owned and managed by Microsoft. It provides a range of capabilities, including software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service (IaaS).

• **Hadoop :** Apache Hadoop is an open-source framework that is suited for processing large data sets on commodity hardware. It is a collection of open source software utilities that facilitates using a network of many computers to solve problems involving massive amounts of data and computation. It provides a software framework for distributed storage and processing of big data using the MapReduce programming model.

• **IBM Cloud :** IBM Cloud is one of the most reliable full-stack cloud platforms, which features a robust cloud computing suite. The cutting-edge platform offers Infrastructure as a Service (IaaS), Platform as a service (PaaS), and Software as a Service (SaaS) to enterprises via the private, hybrid, and public cloud models. The industry-leading services of the IBM Cloud include Analytics, IoT, AI, Blockchain etc.

• **Oracle Cloud :** One of the most trusted cloud platforms Oracle Cloud, offers PaaS, DaaS (Data-as-a-Service), SaaS and IaaS. Oracle Cloud's Infrastructure-as-a-Service includes FastConnect, Ravello, DNS Monitoring, Load Balancing, Database, Networking, Storage, and Compute. The Platform-as-a-Service offerings incorporate Data Management, Business Analytics, Application Development, Cybersecurity, Content Management, and Integration.